

Stokes glacier models, with Firedrake

a finite element tutorial

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github.com/bueler/stokes-ice-tutorial



making this presentation effective

- please ask questions, thanks!
- send questions or comments:
 - by email to elbueler@alaska.edu
 - or by using [issues at github.com/bueler/stokes-ice-tutorial/](https://github.com/bueler/stokes-ice-tutorial/issues)
- these slides are the primary documentation for some Python codes
- please try the codes, and give me your feedback?

Outline

- theory* equations for the glacier dynamics problem
- stage1/ first demo: linear Stokes
- stage2/ power-law Stokes for glacier ice
- stage3/ more power: extruded meshes and 3D glaciers
- theory* evolving glaciers: free-surface equation coupled to Stokes
- stage4/ moving-margin dome solutions
- summary and learning more

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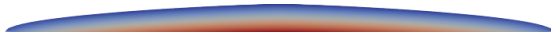
glacier dynamics using a Stokes model: the high-level view

- we apply conservation principles to glaciers:
 - mass conservation ✓
 - momentum conservation ✓
 - energy conservation (not here)
- incompressibility is one aspect of mass conservation
 - another is the free-surface equation ... see `stage4/`
- glaciers are a “slow flow” problem because inertial forces are tiny
 - momentum conservation is therefore called a “stress balance”
 - velocity & pressure are **instantaneously determined** by glacier geometry and boundary stresses

velocity



pressure



- **Stokes equations = stress balance + incompressibility**



George Stokes

Glen-Steinemann-Nye-Stokes equations

- the canonical glacier dynamics equations are from Glen, Steinemann, Nye, and Stokes; the model is isotropic *power-law Stokes*
- the main equations:

$$-\nabla \cdot \tau + \nabla p = \rho \mathbf{g} \quad \text{stress balance}$$

$$\nabla \cdot \mathbf{u} = 0 \quad \text{incompressibility}$$

$$\tau = B_n |\mathbf{Du}|^{(1/n)-1} \mathbf{Du} \quad \text{flow law}$$

- $\mathbf{u}$ is velocity, p is pressure, τ is the deviatoric stress tensor, and $\mathbf{Du} = \frac{1}{2} (\nabla \mathbf{u} + \nabla \mathbf{u}^T)$ is the strain-rate tensor
 - our model has *constant ice density*; incompressibility follows
- constants (*with default values in demos*):
 - $\rho = 910 \text{ kg m}^{-3}$
 - $|\mathbf{g}| = 9.81 \text{ m s}^{-2}$
 - $n = 3$
 - $B_n = 6.8 \times 10^7 \text{ Pa s}^{1/3}$ $\leftarrow$ constant because isothermal here



John Glen



Samuel G. Steinemann
(1923–2016)



John Nye

viscosity, and avoiding unbounded values

- the classical viscosity relationship $\tau = 2\nu D\mathbf{u}$ eliminates τ from the equations, giving a system for $\mathbf{u}$, p
- however, the unmodified power-law viscosity formula yields unbounded values as $D\mathbf{u} \rightarrow 0$, i.e. for vanishing strain rates
 - note that “ $(1/n) - 1$ ” in $\nu = \frac{1}{2}B_n|D\mathbf{u}|^{(1/n)-1}$ is a negative power
- so we also regularize with **small** $\epsilon > 0$ to generate bounded viscosity

power-law Stokes equations with regularized viscosity

$$-\nabla \cdot (2\nu_\epsilon(|D\mathbf{u}|)D\mathbf{u}) + \nabla p = \rho\mathbf{g} \quad \text{stress balance}$$

$$\nabla \cdot \mathbf{u} = 0 \quad \text{incompressibility}$$

$$\nu_\epsilon(|D\mathbf{u}|) = \frac{1}{2}B_n (|D\mathbf{u}|^2 + (\epsilon D_0)^2)^{((1/n)-1)/2} \quad \text{flow law}$$

- we use $\epsilon = 10^{-4}$ and $D_0 = 1 \text{ a}^{-1}$ in the demonstrations

boundary conditions, as simple as possible

- note that $\sigma = 2\nu_\epsilon D\mathbf{u} - pI$ is the **Cauchy stress tensor**
- stress-free top:

$$\sigma \mathbf{n} = (2\nu_\epsilon D\mathbf{u} - pI) \mathbf{n} = \mathbf{0}$$

- atmospheric pressure could be applied at surface . . . no interesting effect
- no-slip base:

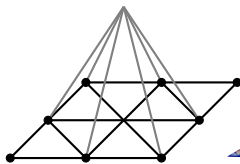
$$\mathbf{u} = \mathbf{0}$$

- **no attempt to model basal sliding** in this tutorial



finite elements ... with no details

- by-hand solution of the Stokes model is impossible for most geometries
 - *exercise*. solve slab-on-a-slope, which is an exception
- so we solve numerically using the finite element (FE) method
- the FE method allow us to start from a mesh of triangles (or quadrilaterals, prisms, ...) over our domain, then express the approximate solution in terms of functions built from the mesh, but I will skip the details ...



- the whole point of Firedrake is **suppress the details** of FE!
 - but try Elman, Sylvester & Wathen (2014) for self-study?
- however, one needs to know a key step: **write a weak form**

the weak form

- start from the equations, now called the *strong form*:

$$-\nabla \cdot (2\nu_\epsilon(|D\mathbf{u}|) D\mathbf{u}) + \nabla p = \rho \mathbf{g} \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0 \quad (2)$$

- multiply (1) by test velocity $\mathbf{v}$ and (2) by test pressure q , integrate by parts, and combine into one **nonlinear residual function**:

$$F(\mathbf{u}, p; \mathbf{v}, q) = \int_{\Lambda} 2\nu_\epsilon(|D\mathbf{u}|) D\mathbf{u} : D\mathbf{v} - p \nabla \cdot \mathbf{v} - q \nabla \cdot \mathbf{u} - \rho \mathbf{g} \cdot \mathbf{v} \, dx$$

- Λ is the spatial domain of ice
 - “ dx ” = $dx \, dy \, dz$ in 3D, and “ dx ” = $dx \, dz$ in the planar case
- the actual **weak-form problem statement** is “residual = 0”:

$$F(\mathbf{u}, p; \mathbf{v}, q) = 0 \quad \text{for all test functions } \mathbf{v} \text{ and } q$$

- the FE method solves the same problem, but over a finite-dimensional space

- nonlinear residual function from last slide:

$$F(\mathbf{u}, p; \mathbf{v}, q) = \int_{\Lambda} 2\nu_{\epsilon}(|D\mathbf{u}|) D\mathbf{u} : D\mathbf{v} - p\nabla \cdot \mathbf{v} - q\nabla \cdot \mathbf{u} - \rho \mathbf{g} \cdot \mathbf{v} \, dx$$

$$\text{with } \nu_{\epsilon}(|D\mathbf{u}|) = \frac{1}{2} B_n (|D\mathbf{u}|^2 + (\epsilon D_0)^2)^{((1/n)-1)/2}$$

- we write this using the *Unified Form Language* (UFL) in Firedrake:

```
def D(w):  
    return 0.5 * (grad(w) + grad(w).T)  
  
Du2 = 0.5 * inner(D(u), D(u)) + (eps * Dtyp)**2.0  
nu = 0.5 * Bn * Du2**((1.0 / n - 1.0)/2.0)  
  
fbody = rho * Constant((0.0, 0.0, -g))  
F = ( 2.0 * nu * inner(D(u), D(v)) \  
      - p * div(v) - q * div(u) - inner(fbody, v) ) * dx
```

- Python UFL code is quite similar to the math

mixed finite elements for fluid problems

- other details we need to know about:
 1. choose **separate function spaces** for the velocity and the pressure
 2. choose finite element spaces that (the experts say) are **stable**
- Firedrake makes choosing such **mixed elements** easy:

```
V = VectorFunctionSpace(mesh, 'CG', 2) # u space is P2
W = FunctionSpace(mesh, 'CG', 1)      # p space is P1
Z = V * W                             # mixed space
up = Function(Z)
u, p = split(up)
v, q = TestFunctions(Z)
```

- this $P_2 \times P_1$ choice is called Taylor-Hood
- other mixed spaces are just as easy
- we choose $P_2 \times DQ_1$ in stage4/, for reasons ...

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github.com/bueler/stokes-ice-tutorial



the linear Stokes equations

- if we make viscosity constant ($\nu_0 > 0$) then we get a linear Stokes system:

$$\begin{aligned} -\nabla \cdot (2\nu_0 D\mathbf{u}) + \nabla p &= \rho \mathbf{g} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned}$$

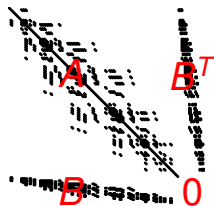
- classical reformat with $2\nabla \cdot D\mathbf{u} = \nabla^2 \mathbf{u}$:

$$\begin{aligned} -\nu_0 \nabla^2 \mathbf{u} + \nabla p &= \rho \mathbf{g} \\ -\nabla \cdot \mathbf{u} &= 0 \end{aligned}$$

- which has symmetric block structure:

$$\begin{bmatrix} A & B^\top \\ B & 0 \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ p \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix}$$

- $A = -\nu_0 \nabla^2 \geq 0$
- $B^\top = (-\nabla \cdot)^\top = \nabla$ (integration-by-parts)



running the demonstration programs

- let's run a demo
- install Firedrake following these directions:
firedrakeproject.org/install.html
- activate the Python virtual environment every time you use Firedrake:

```
$ source ~/venv-firedrake/bin/activate
```

- other tools you will need:
 - Gmsh gmsh.info
 - Paraview paraview.org
- clone this tutorial, both for the codes and these slides:

```
$ git clone https://github.com/bueler/stokes-ice-tutorial.git
```



- *purpose*: solve linear Stokes on a trapezoidal “glacier”
- *source files*: `trap.geo`, `solve.py` ← inspect these
- *generated files*: `trap.msh`, `trap.pvd`

```
$ cd stage1/  
$ gmsh -2 trap.geo           # mesh the domain  
$ gmsh trap.msh              # view the mesh  
$ python3 solve.py           # solve linear Stokes  
$ paraview trap.pvd          # view the solution
```



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Newton's method

- linear Stokes in `stage1/` solves a single linear system:

$$\begin{bmatrix} A & B^T \\ B & 0 \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ p \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix}$$

- default solver: Firedrake $\rightarrow$ PETSc KSP $\rightarrow$ MUMPS
- but Glen-Stokes residual function is **nonlinear in $\mathbf{u}$** :

$$F(\mathbf{u}, p; \mathbf{v}, q) = \int_{\Lambda} 2 \nu_{\epsilon}(|D\mathbf{u}|) D\mathbf{u} : D\mathbf{v} - p \nabla \cdot \mathbf{v} - q \nabla \cdot \mathbf{u} - \rho \mathbf{g} \cdot \mathbf{v} \, dx$$

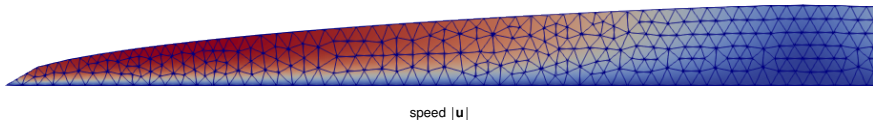
- fortunately, Firedrake applies Newton's method by default if we write the problem as `solve(F == 0, ...)`
- what you need to know about Newton's iteration:
 - it does linearization around each iterate
 - Firedrake uses UFL symbolic differentiation for linearization
 - Firedrake asks PETSc's SNES to do the Newton iteration
 - SNES options will monitor and control the iteration



Isaac Newton

- *purpose*: solve the power-law Stokes equations on a dome-shaped glacier with a flat bed
- *source files*: `domain.py`, `solve.py` ← inspect these
- *generated files*: `dome.geo`, `dome.msh`, `dome.pvd`

```
$ cd stage2/  
$ python3 domain.py      # generate domain outline  
$ gmsh -2 dome.geo       # mesh the domain  
$ python3 solve.py       # solve power-law Stokes  
$ paraview dome.pvd      # view the solution
```



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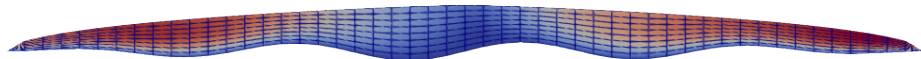
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a more practical and powerful glacier solver

- this stage takes a more practical and powerful approach
- we switch to an **extruded mesh**, and allow a bumpy bed (below)
 - the **base mesh** has lower dimension
 - now the elements are quadrilaterals in the planar case, versus triangles in stage2/
- we demonstrate (mostly) **dimension-independent programming**
 - code for 2D and 3D glacier geometry is nearly the same; elements are prisms in 3D
 - dimension chosen at runtime
- additional bells & whistles:
 - rescaling the equations
 - vertical grid sequencing, with over-regularization on coarse meshes
 - additional diagnostic fields in the output `.pvd` file

solution speed $|\mathbf{u}|$ for default run:



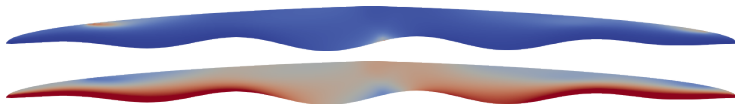
- *purpose*: a practical and robust solver
 - construct mesh by vertically extruding a base mesh
 - rescale the equations
 - vertical grid sequencing
 - write ν_ϵ , τ , $p - p_{\text{hydrostatic}}$ into output file
- *source file*: `solve.py`
- *generated file*: `dome.pvd`

practical geometry
better conditioning
better initial iterates
better diagnostics

← inspect this

```
$ cd stage3/  
$ python3 solve.py                # default resolution  
$ python3 solve.py -mx 400 -refine 2 # higher resolution  
$ paraview dome.pvd
```

effective viscosity ν_ϵ and deviatoric stress magnitude $|\tau|$:



evidence of convergence

- we can now set resolution at runtime, for example:

```
$ python3 solve.py -b0 0 -mx 400 -mz 4 -refine 2
```

- using optional flat bed (`-b0 0`), to compare results to `stage2/`
 - initial mesh has 400 elements in horizontal and 4 in vertical
 - refine the vertical twice (by factors of four), to generate a 400×64 mesh
- velocity results, in meters per year, from a refinement study:

mesh	$\Delta x \times \Delta z$ (m)	average $ \mathbf{u} $	maximum $ \mathbf{u} $
50×8	400×125	1782	3263
100×16	200×63	1766	3211
200×32	100×31	1761	3196
400×64	50×16	1759	3191
800×128	25×8	1759	3190

- velocities are apparently converging, but we do not know the exact solution
... this is not verification

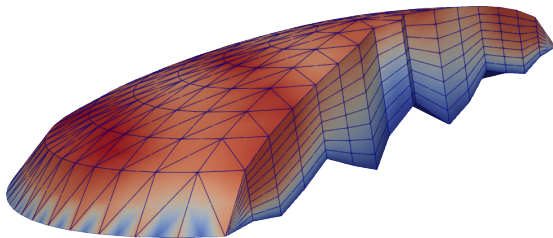
- as usual with Firedrake, everything works in parallel
 - extruded mesh gets partitioned following base mesh partition
 - see `rank` field in `.pvd` output; result with 4 processes:



- using the default direct solver (Mumps), results are essentially independent of number of processes
 - however, speed-up is sublinear because of significant serial workloads
- ```
$ python3 solve.py -mx 800 -refine 2 # 38 secs run time
```
- ```
$ mpiexec -n 4 python3 solve.py -mx 800 -refine 2 # 20 secs run time
```
- your performance results may vary ...

- *purpose*: demonstrate a (mostly) dimension-independent Firedrake code, to model a 3D glacier
- *source file*: `solve.py` ← inspect this
- *generated file*: `dome.pvd`

```
$ python3 solve.py -dim 3 -baserefine 3 -refine 1  
$ paraview dome.pvd
```



3× vertical exaggeration, coloring by speed $|\mathbf{u}|$

- this run is about as far as I can refine on my office desktop:

```
$ mpiexec -n 12 python3 solve.py -dim 3 -baserefine 4 \  
    -refine 2 -s_snes_atol 1.0e-2 -o finest.pvd
```

- *limiting factor*: the **direct solver** for each Newton step linear system uses too much memory when generating the LU factors
 - direct solvers experience *fill-in*, especially in 3D; see B (2021b)
- for higher resolution we need a **scalable solver**
 - such a solver would be built by composition of features: matrix-free, Schur-complement preconditioning, multigrid on the **u-u** block; see B (2021b)
 - for a scalable implementation see Isaac, Stadler, & Ghattas (2015)
 - addressing scalability properly is a different talk, but see B (2023)

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but don't glaciers actually change with the climate?

- we do not yet have an evolving glacier model
 - time derivatives have not even appeared!
 - the climate (e.g. surface mass balance) has not appeared as an input

Q. how do glaciers change shape?

A. via the **free-surface equation**, a kinematical and mass conservation statement:

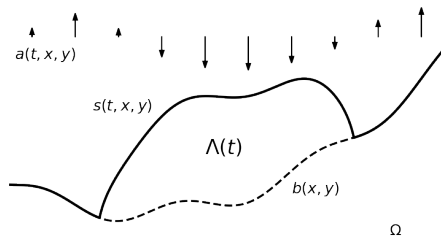
$$\frac{\partial s}{\partial t} - \mathbf{u}|_s \cdot \mathbf{n}_s = a$$

- $z = s(t, x, y)$ is the surface elevation of the glacier
 - $\mathbf{u}|_s$ is the surface value (*trace*) of the ice flow velocity
 - $\mathbf{n}_s = \left\langle -\frac{\partial s}{\partial x}, -\frac{\partial s}{\partial y}, 1 \right\rangle$ is a normal vector to the surface
 - $a(t, x, y)$ is the **surface mass balance**, the source term
- equation holds on $(0, T) \times \Omega$, where $\Omega \subset \mathbb{R}^d$ is a fixed map-plane region
 - initial surface $s(0, x, y)$ is needed

coupled free-surface & power-law Stokes equations

$$\begin{aligned}\frac{\partial s}{\partial t} - \mathbf{u}|_s \cdot \mathbf{n}_s &= a && \text{free-surface equation} \\ -\nabla \cdot (2\nu_\epsilon D\mathbf{u}) + \nabla p &= \rho \mathbf{g} && \text{stress balance} \\ \nabla \cdot \mathbf{u} &= 0 && \text{incompressibility}\end{aligned}$$

- 3 solution variables: s , $\mathbf{u}$, p
- 2 coupling directions:
 - I. the free-surface equation has $\mathbf{u}|_s$ as a coefficient
 - II. $s(t, x, y)$ determines the evolving domain $\Lambda(t)$ on which the Stokes problem is solved at each instant



weak form of coupled system of equations

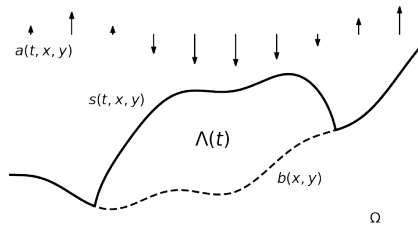
- we get the weak form of the coupled system in the usual way
- multiply by test functions and integrate:

$$\int_{\Omega} \left(\frac{\partial s}{\partial t} - \mathbf{u}|_s \cdot \mathbf{n}_s - a \right) r \, dx = 0 \quad \forall r$$

$$\int_{\Lambda(t)} 2 \nu_{\epsilon} (|D\mathbf{u}|) D\mathbf{u} : D\mathbf{v} - p \nabla \cdot \mathbf{v} - \rho \mathbf{g} \cdot \mathbf{v} \, dx = 0 \quad \forall \mathbf{v}$$

$$\int_{\Lambda(t)} -(\nabla \cdot \mathbf{u}) q \, dx = 0 \quad \forall q$$

- high-level choices for system:
 1. spaces and elements for $s, \mathbf{u}, p$?
 2. time-stepping: explicit or implicit?
 3. solved together (*monolithic*) or split?



challenges in solving the coupled system

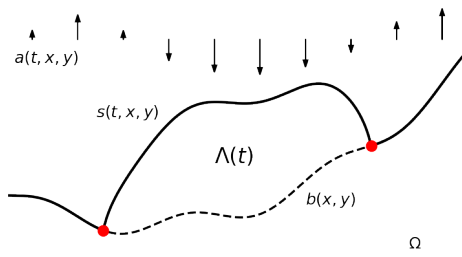
- early/naive approaches to solving this coupled problem have been fragile
- the method everyone tries first is to solve power-law Stokes for the current geometry, then advance the surface, and repeat
 - this is an *explicit* approach, which *splits* the coupling
 - resulting performance is **stability-limited**, in a manner which is not well understood
- what are the underlying challenges in designing better schemes?
- ① the mesh used for the $\mathbf{u}, p$ solver, or at least its geometry, changes at every time step
 - this makes monolithic and/or implicit implementations difficult
- ② the coupled problem mixes advective and diffusive aspects
 - proportions depending on stress boundary conditions
 - precise time-step restrictions remain unclear
 - expected performance scaling remains unclear (B 2023)
 - but `stage4/` applies recent theoretical progress!

challenges in solving the coupled system

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also, glaciers have moving margins!

- the **surface elevation must be above the bed**
- enforcing this obvious fact provides the (lateral) free-boundary condition for a land-based glacier
 - the weak form is a **variational inequality** (B 2021b, Chapter 12)
 - the thickness and flux are both zero at the free boundary
- solving the coupled problem subject to this condition determines the location of the moving glacier margin (free boundary)



Outline

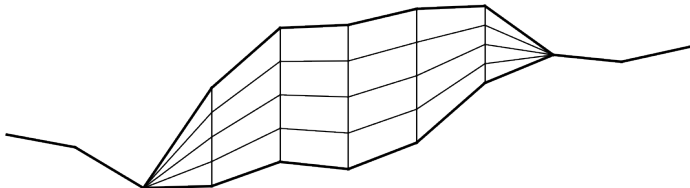
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evolving extruded meshes

- we will demonstrate and test a 2D solver which is **semi-implicit**, **split**, **stabilized**, and **moving-margin**
 - ... as explained on the next few slides
 - caution: “stabilized” will mean several things here
- using an extruded mesh
- between continuous P_1 surfaces $s = s(t, x)$ and $b = b(t, x)$:
 - the Stokes domain $\Lambda(t)$ is **polygonal** (in space) at each t
 - the elements are quadrilaterals (in 2D)



- our solver is based on these **3 new techniques** which modify the weak form of the coupled problem:
 1. first-order, and mostly explicit, time-stepping
 - A. but we add an implicit **edge stabilization** in the free-surface equation
 - B. and we determine the time step by a CFL condition
 2. split, with alternating free-surface and Stokes solves
 - but we add a **load stabilization** coupling to the Stokes solver
 3. the coupled free-surface problem is solved as a **variational inequality** which determines the glacier extent and margin
 - A. we apply a bounds-respecting Newton method
 - B. we allow variable ice extent on the extruded mesh
 - C. we prefer pressure elements with interior degrees of freedom
- **the next slides give citations and details on these techniques**
 - techniques 1 and 2 are the “Swedish stabilizations”

technique 1A: implicit edge stabilization in free-surface equation

- this technique is from Tominec et al (2026)
 - see also Ern & Guermond (2013), and refs therein
- add FE **edge stabilization**, applied semi-implicitly:

$$\int_{\Omega^h} \left(\frac{s_{k+1}^h - s_k^h}{\Delta t} - \mathbf{u}|_{s_k^h} \cdot \mathbf{n}_{s_k^h} - a \right) r^h dx + J(s_{k+1}^h, r^h) = 0$$

for all test functions r^h , where

$$J(s, r) = \sum_{e \in E^h} \int_e \frac{1}{2} h_e^2 |\mathbf{u}|_{s_k^h}| \llbracket \nabla s \rrbracket \llbracket \nabla r \rrbracket dS$$

- s_k^h is the FE approximation of surface elevation at time t_k
- the system to solve for unknown s_{k+1}^h is **linear**
- E^h is the set of interior edges in the triangulation Ω^h
- $\llbracket \mathbf{v} \rrbracket = \mathbf{v}_+ \cdot \mathbf{n}_+ - \mathbf{v}_- \cdot \mathbf{n}_-$ is a **vector field jump along an edge**
- h_e is the average of the diameters of the two cells which meet e
- note surface velocity ($\mathbf{u}|_{s_k^h}$) and slope ($\mathbf{n}_{s_k^h}$) are evaluated explicitly, at t_k



Igor Tominec



Lukas Lundgren



Andres Lofgren



Josefin Ahlstrom

technique 1B: CFL determination of time step

- our application adds an advancing glacier margin, which is not considered in the stability proof from Tominec et al (2026)
- edge stabilization technique 1A applies split and semi-implicitly, with **velocity evaluated explicitly**
 - thus margin advance is limited to one base-mesh element per time step
- the time step Δt is therefore limited by a CFL criterion:

$$\Delta t = C_{\text{CFL}} \min \frac{h}{|\mathbf{u}|}$$

- we get reasonable results with $C_{\text{CFL}} = 0.25$
 - Ern & Guermond (2013) use $C_{\text{CFL}} = 0.2$ or 0.25 on linear advections with weighted edge stabilization (and SSP3, RK4 schemes)

technique 2: semi-implicit coupling heuristic in stress balance

- from Löfgren et al (2022), as refined by Tominec et al (2026)
- add **extrapolated surface mass balance load** to Stokes form:

$$\int_{\Lambda(t_k)} 2\nu_\epsilon(|D\mathbf{u}|) D\mathbf{u} : D\mathbf{v} - p\nabla \cdot \mathbf{v} - \rho \mathbf{g} \cdot \mathbf{v} dx \\ + \int_{\Gamma_s(t_k)} \rho g \Delta t \left[\frac{1}{2} (\mathbf{u}|_{s_k} \cdot \mathbf{n}_{s_k}) + a \right] (\mathbf{v} \cdot \mathbf{n}) ds = 0$$

- $\Lambda(t_k)$ is the Stokes domain defined by the current surface elevation s_k
 - $\Gamma_s(t_k)$ is the upper surface of $\Lambda(t_k)$
 - $\mathbf{n}_{s_k} = \langle -\nabla s_k, 1 \rangle$ is an outward normal on $\Gamma_s(t_k)$
 - $\mathbf{n} = \mathbf{n}_{s_k} / |\mathbf{n}_{s_k}|$ is the outward *unit* normal
- concept: modify the stress balance to stabilize the coupled solve
 - Reynold's transport theorem motivates this (symmetrized) extrapolation
 - ... which puts time step Δt and SMB a inside the Stokes solve!
 - Tominec et al (2026) prove that the L^2 norm of the evolving surface is **controlled**; it can't blow up



Andres Löfgren



Josefin Ahlkrona



Christian Helanow

- *purpose*: exhibit **numerical stability** and **exact mass conservation** for an evolving surface between lubricated walls
 - the geometry considered by Löfgren et al (2022), Tominec et al (2026)
- *source files*: `geometry.py`, `halfar.py`, `solve.py` ← **inspect these**
- *generated file*: `dome.pvd`

```
$ cd stage4/  
$ python3 solve.py -walls -L 8000 -omovie.pvd  
$ paraview movie.pvd
```



instability showing need for stabilization technique 2

- *without* technique 2, the maximum size of the stable time step is significantly restricted (Löfgren et al 2022)
 - many time step choices produce “sloshing” and Stokes solver failures
 - e.g. runs to produce table: `$ python3 solve.py -walls -L 8000 -noload -nocfl -mx MX -maxdt DT`
with $MX = 25, 50, 100, 200, 400$
- to demonstrate what happens *without* technique 2, for each resolution in this table we started with $\Delta t = 1$ a, and reduced Δt by a factor of two until the 5 a run completes without a solver failure

Δx (m)	stable Δt (a) w/o technique 2
640	0.25
320	0.25
160	0.125
80	0.125
40	0.125

- Tominec et al (2026) does additional experimentation
- with technique 2, arbitrary time steps are stable, and exhibit no sloshing

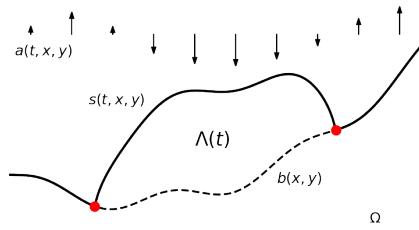
technique 3A: semi-implicit variational inequality for free surface

- define the FE **admissible set** $\mathcal{K}^h = \{r^h \in P_1 : r^h \geq b^h\}$
 - the P_1 bed elevation function b^h is time-independent
- the new surface $s_{k+1}^h \in \mathcal{K}^h$ solves this variational inequality weak form:

$$\int_{\Omega} \left(\frac{s_{k+1}^h - s_k^h}{\Delta t} - \mathbf{u}|_{s_k^h} \cdot \mathbf{n}_{s_k^h} - a \right) (r^h - s_{k+1}^h) dx + J(s_{k+1}^h, r^h - s_{k+1}^h) \geq 0$$

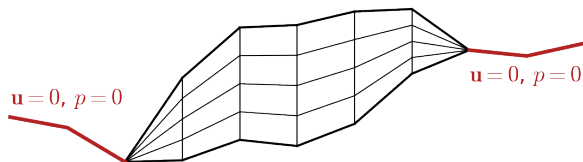
for all r^h in $\mathcal{K}^h$

- includes edge stabilization
- velocity $\mathbf{u}|_{s_k^h}$ still explicit**
- the computed margin location is where s_{k+1}^h meets b^h
- we apply the `vinewtonrsls` bounds-respecting Newton solver from PETSc (B 2021b)



technique 3B: extruded mesh with variable ice extent

- the base mesh covers all of Ω , including areas with no ice
- the extruded mesh has a fixed number m_z of levels in the vertical
 - $\mathbf{u}, p$ are defined at each node of the extruded mesh
 - our meshes and fields have **constant memory footprint** as we time-step
- in columns with no ice we lock/pinch $\mathbf{u}, p$ values to zero



- s_k^h, b^h define ice domain $\Lambda(t_k)$ in this semi-implicit framework
- $s_k^h = b^h$ at a base mesh node $\implies$ no ice in that extruded column
- `PinchColumnPressure|Velocity` **sub-class** `DirichletBC`

technique 3C: pressure elements with interior degrees of freedom

- however, we do **not** want to pinch p to be zero anywhere in the closest-to-margin elements containing ice
 - doing so causes an unintended Dirichlet condition on the pressure
- therefore prefer pressure elements with interior degrees of freedom:

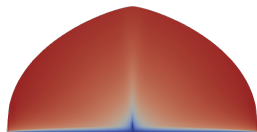
```
W = FunctionSpace(mesh, "DQ", 1, variant="spectral") # best
```

```
W = FunctionSpace(mesh, "DG", 0) # adequate
```

- for such elements the moving-margin mass conservation error (later slides) is provably one-sided

- *purpose*: moving-margin free surface, exhibiting numerical stability
- *source files*: `geometry.py`, `halfar.py`, `solve.py` ← inspect these
- *generated file*: `movie.pvd`

```
$ python3 solve.py -mx 200 -mz 10 -T 100 -L 20000 \  
    -omovie movie.pvd  
$ paraview movie.pvd
```

 $t = 0$ a $t = 100$ a

10× vertical exaggeration, $\log(|\mathbf{u}|)$ coloring

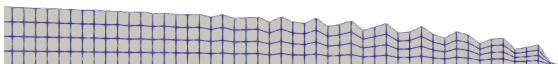
instability showing need for stabilization technique 1A

- especially near the glacier margin, implicit **edge stabilization** (technique 1A) **is clearly needed**
- compare these runs:

```
$ python3 solve.py -mx 200 -T 1
```



```
$ python3 solve.py -mx 200 -T 1 -noedge
```



- runs with `-walls` generally do not show the need

instability showing need for stabilization technique 1B

- need for technique 1B (CFL for time-step length) is more subtle
- compare the margin shape at the end of these two runs:

```
$ python3 solve.py -mx 100
```



```
$ python3 solve.py -mx 100 -nocfl -maxdt 1.0
```



- advancing margin shape without CFL-limited time steps is wrong
 - bulging and not downsloping
 - not close to Halfar margin location (next slides)
 - but not obviously unstable

Q. is our time-dependent solver correct?

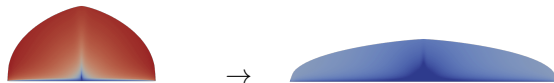
- it is not clear how to build a moving-margin, coupled Stokes-surface exact solution
- however, such exact solutions exist for **shallow ice approximation (SIA)**
- SIA exact solutions are **close** to solving the free-surface + Stokes coupled equations when the aspect ratio $\epsilon = (\text{height})/(\text{width})$ is small
- next slide: we compare with the Halfar (1981) exact solution
 - on a **shallow initial dome geometry**, SIA is reasonably accurate
 - e.g. initial height 1000 m and width 20 km yields $\epsilon = 0.05$



Peter Halfar

evidence of correctness for a shallow, moving-margin dome

- consider runs of 10 years at different horizontal resolutions:
 - `$ python3 solve.py -mz 10 -T 10 -mx MX`
 - use $MX = 50, 100, 200, 400, 800$, giving $75 \leq \Delta x \leq 1200$ m
 - from 33 ($MX = 50$) to 557 ($MX = 800$) adaptive-scheme time steps
 - max computed ice speed 4600 m/a initially; 160 m/a for final state



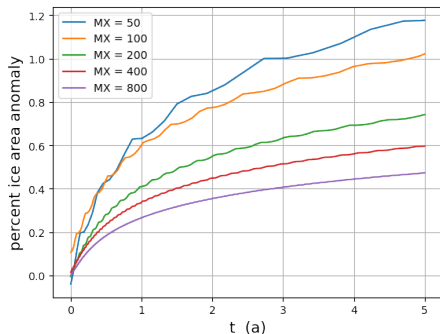
- compare final computed geometry to Halfar's prediction:

Δx (m)	width (km)	center height (m)
600	28.800	697.789
300	28.800	697.026
150	28.800	696.107
75	28.950	695.670
37.5	28.875	695.329
Halfar	28.993	689.829

- reasonably close, but we do *not* expect convergence to Halfar value

mass conservation errors with moving glacier margins

- since $a = 0$ here, mass is exactly conserved in the continuum model
- unfortunately, our scheme **does not conserve mass** here
- numerical mass (= ice area) drifts about 1% over 5 years
 - curves should be horizontal lines
 - higher res $\implies$ better conserve
- this is **only a moving margin effect**
 - our scheme conserves mass for fixed walls, as proved by Tominec 2026
 - here mass is non-decreasing; prove from VI using $a = 0$
 - this margin mass conservation issue is distinct from those in Jarosch et al (2013) and B (2021a)



Outline

- theory* equations for the glacier dynamics problem
- stage1/ first demo: linear Stokes
- stage2/ power-law Stokes for glacier ice
- stage3/ more power: extruded meshes and 3D glaciers
- theory* evolving glaciers: free-surface equation coupled to Stokes
- stage4/ moving-margin dome solutions
- summary and learning more

github.com/bueler/stokes-ice-tutorial



summary and outlook

- Firedrake makes solving the glacier Stokes problem **easy for the new numerical modeler**:
 - the Python codes in this tutorial are short
 - UFL (weak forms) and PETSc (nonlinear solvers) are built-in to the stack
- **modern techniques work**:
 - extruded meshes (Firedrake), because glaciers are thin flows
 - Swedish stabilizations, so semi-implicit time-stepping is safe and easy
 - Tominec et al (2026) has 1st coupled Stokes-surface *proof* (to my knowledge)
 - bounds-constrained Newton solvers (PETSc), giving moving margins
- remaining challenges and improvements:
 - 1 make the Stokes solver faster by replacing the direct linear algebra with a **scalable matrix-free solver** (presumably multigrid)
 - 2 make the simulation faster by combining the free-surface equation and the Stokes solver in a **monolithic implicit step** solved by a scalable solve (presumably multigrid = FASCD?)
 - 3 make time-stepping glacier simulations **mass conserving even in free-boundary cases**

github.com/bueler/stokes-ice-tutorial



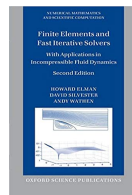
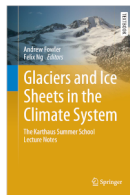
- Firedrake: firedrakeproject.org
- PETSc: petsc.org



Firedrake

 PETSc

- for power-law Stokes eqns, see Ch. 1 by Hewitt in Fowler & Ng (2021)
- for finite elements and Stokes: Elman, Silvester, & Wathen (2014)
- for PETSc, Firedrake, and Stokes: Bueler (2021b)



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